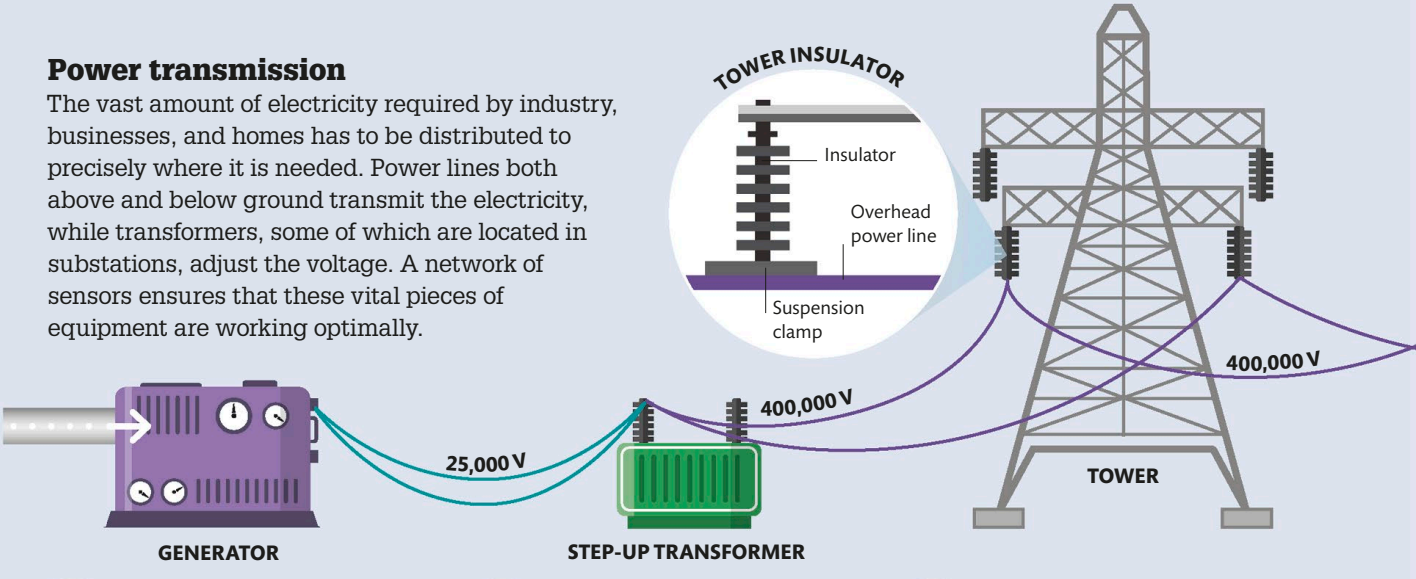
Electricity supply

Most electricity is generated at large power stations (see pp.20–21) and then distributed to consumers, such as factories and homes, sometimes over substantial distances. This involves a large and complex network of cables and facilities together known as a power grid.

Power transmission

The vast amount of electricity required by industry, businesses, and homes has to be distributed to precisely where it is needed. Power lines both above and below ground transmit the electricity, while transformers, some of which are located in substations, adjust the voltage. A network of sensors ensures that these vital pieces of equipment are working optimally.



Transmission towers
Transmission towers are usually tall steel and aluminum towers featuring a lattice or tubular frame. They carry power lines at safe heights above ground level and feature insulators to separate the high-voltage cable from the earthed tower.

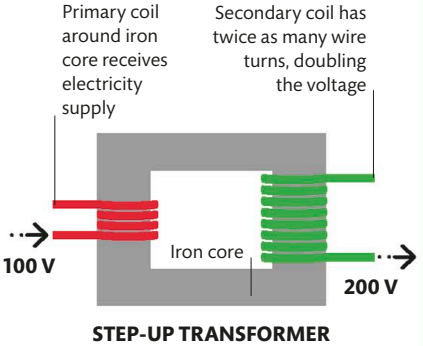
1 Power station
A generator located in a power station converts kinetic energy into electrical energy. This provides a supply of alternating current (AC) electricity (see p.16) typically at 25,000 volts.

2 Grid substation
A grid substation uses transformers to increase the voltage, typically to 400,000 volts. The higher the voltage, the less energy is lost as heat from resistance when the electricity travels along power lines.

3 High-voltage tower lines
Transmission towers are often made from steel-reinforced aluminum. They are fitted with glass or ceramic insulators to prevent electricity from the power lines traveling down the steel tower to the ground.

TRANSFORMERS

A transformer alters the voltage of electricity through the process of electromagnetic induction. First, alternating current (AC) flows through a transformer's primary coil, wound around an iron core. It produces a changing magnetic field, which induces a voltage in a secondary coil. If the secondary coil contains more wire than the primary, the voltage is increased, or stepped up. Less wire results in a voltage decrease, or step down.



STEP-UP TRANSFORMER

HOW CAN BIRDS PERCH ON POWER LINES?

Electricity always flows along the path of least resistance. Birds do not conduct electricity well, so the electricity bypasses them, instead continuing to travel along the power line.